Transferable Operational Failure Priors for Coding Agents

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Abstract

Coding agents fail in predictable ways, and we ask whether that predictability transfers between agents. Using a corpus of 75,933 actions from four harnesses (Claude Code, Codex, Cline, OpenCode) across three machines, we test whether action-kind failure rates learned from one agent predict another agent's failures. Twelve directional source→target comparisons are evaluated by session-level bootstrap — the correct inferential unit, since actions cluster within sessions — with family-wise correction over the twelve tests (Holm). **No pair shows positive transfer.** The only three comparisons to survive Holm are significantly *below* chance (AUC 0.394, 0.401, 0.416), i.e. the borrowed prior misleads rather than merely failing to help. The single pair that appeared to transfer in our earlier analysis, cline→opencode (0.534, raw $p = 0.009$), does not survive Holm; it would need $p \leq 0.006$. Under the less conservative Benjamini–Hochberg procedure it does survive, as one of six, the other five being anti-predictive — so we report the positive result as BH-dependent and not established. We also show that the choice of bootstrap unit is a first-class methodological variable here: the incorrect action-level bootstrap makes two pairs look like positive transfer where the correct session-level unit finds none. On the positive side, we can report one leak-free result: our earlier *cold-start* claim was contaminated by train/test leakage and has been withdrawn and re-run. In the corrected experiment — priors built only on a held-out training pool, scored on a fixed held-out test set, paired intervals — the borrowed prior sits *below* chance at every history size (0.483) and never contributes, while two training sessions of local history already reach 0.600 and beat the borrowed prior by +0.114. There is no regime in which borrowing helps, which is a stronger claim than the one we originally made. We report the negative result, and the anti-predictive result, as the primary findings.

1 Introduction

Large language model-based coding agents (Claude Code, Codex, Cline, OpenCode) execute hundreds of tool calls per session. A small fraction of these calls fail. Failure is not uniformly distributed across action kinds: some (e.g., `bash:cd`, `read`) almost never fail, while others (e.g., `bash:ssh`, `exec`) fail at rates an order of magnitude higher. This structure is *operational*: it belongs to the agent's interaction with its environment, not to the code being written.

We ask: can a failure prior learned from one agent predict another agent's failures? This is a *transfer* question. The practical motivation is cold-start: a new agent on a new machine has no local history, and a borrowed prior from a different agent's accumulated record may be better than nothing.

The question has a clear failure mode. If failure structure is primarily a property of the user, the repository, or the task, then cross-agent transfer will be at chance. If it is a property of the agent's tool-use patterns, then transfer will succeed wherever the two agents' action-kind vocabularies overlap.

2 Data

We use a redacted corpus of 75,933 coding-agent actions extracted from 402 sessions across three machines (`pansa`, `omo`, `feather`) and four harnesses (Table 1). Targets are replaced by salted digests; no recorded command is readable. The label is `marker_error`: whether the tool output contained a failure marker (Traceback, `error[E0]`, FAILED, fatal:, panicked at, command not found).

Harness	Sessions	Actions	Error rate
OpenCode	288	48,460	0.048
Claude Code	42	14,457	0.061
Cline	69	12,354	0.074
Codex	3	662	0.038
Merged	**402**	**75,933**	**0.055**

Table 1: Action corpus. Error rate is per-action marker_error rate. Sessions are deduplicated across machines (188 duplicate session records were dropped).

The four labels are not the same measurement: Codex's is a real exit code, OpenCode's and Claude Code's are harness judgements, Cline's is mostly output markers. Cross-harness comparisons therefore mix a small amount of label drift into every number.

3 Method

For each harness h, we compute per-action-kind failure rates:

$$\hat{p}_h(k) = \frac{\text{failures of kind } k \text{ in } h}{\text{total actions of kind } k \text{ in } h}.$$

For cross-harness transfer from source s to target t, we predict each target action's failure probability using the source's rates:

$$\hat{p}(a) = \begin{cases} \hat{p}_s(\text{kind}(a)) & \text{if kind}(a) \in \text{source vocabulary} \\ 0.5 & \text{otherwise (chance for unseen kinds).} \end{cases}$$

We score predictions against target t's labels using the area under the receiver operating characteristic curve (AUC). A bootstrap over target sessions (500 resamples) provides confidence intervals.

For the in-corpus baseline, we use leave-one-session-out: fit $\hat{p}_h(k)$ on all sessions but one, predict the held-out session.

4 Results

4.1 Cross-harness transfer

Figure 1 shows the full transfer AUC matrix. Every off-diagonal entry is a source→target transfer; the diagonal is in-corpus (leave-one-session-out).

The inferential unit. All intervals below bootstrap over *sessions*, not actions. Actions are clustered within sessions (median 101 actions per session), so an action-level bootstrap treats correlated observations as independent and narrows the interval artificially. The effect is smaller than

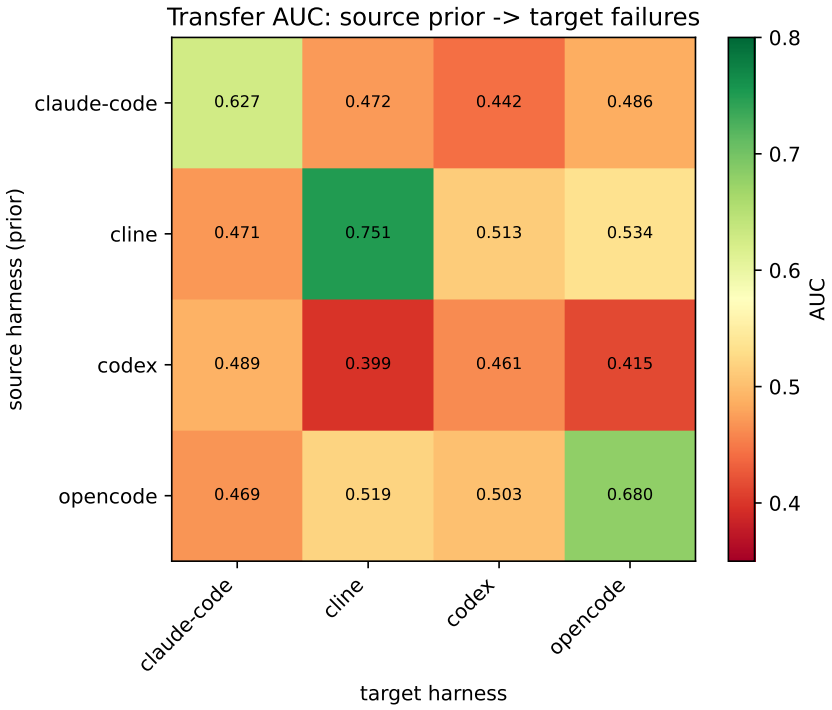


Figure 1: Cross-harness transfer AUC matrix. Each cell shows the AUC of predicting target t's failures using source s's per-kind rates. Diagonal entries are in-corpus (leave-one-session-out). Only one of the twelve off-diagonal entries has a session-bootstrap interval excluding chance (cline→opencode, 0.534 [0.509, 0.559]); the other eleven intervals include 0.5. Even the surviving pair remains well short of opencode's in-corpus baseline (0.680 [0.642, 0.718]).

we first assumed but real: under an action-level bootstrap two of twelve pairs have intervals excluding chance, while under the session-level bootstrap one does. The pair that changes status is opencode→cline, which moves from 0.519 [0.505, 0.534] to 0.519 [0.468, 0.573] — the point estimate is identical and only the interval width changes, which is precisely the failure mode. We report session-level intervals throughout, and the recomputation is stored in `research/bootstrap_unit_check.json`.

4.2 Multiplicity correction

Twelve directional tests are reported. Selecting the smallest p from twelve and quoting its interval is not evidence of an effect; the interval does not account for the selection. We therefore apply family-wise correction over the twelve comparisons, on session-level bootstrap p-values from 10,000 replicates.

No pair shows positive transfer after family-wise correction. The three comparisons that survive Holm are all significantly *below* chance (0.394, 0.401, 0.416) — the borrowed prior is anti-predictive, not merely uninformative. All three involve Codex, as source in two cases and target in one; we do not have a mechanism for that and flag it as an observation, not a finding.

The pair that appeared to transfer in our earlier analysis, cline→opencode (0.534), is *nominally* significant only. Its Holm threshold is set by its rank among the twelve: it sits at rank 5, so the denominator is $m - r + 1 = 12 - 5 + 1 = 8$ and the threshold is $0.05/8 = 0.00625$. Its nominal $p = 0.0094$ exceeds that, so it does *not* survive Holm. We use "nominally significant" for unadjusted $p < 0.05$ throughout and reserve "survives Holm" for corrected results; six pairs are

Table 2: All twelve directional comparisons, session-level bootstrap (10,000 replicates), ordered by raw p. The AUC column is the *bootstrap mean*, which differs from the point estimate for pairs with very wide intervals (see the note below the table). 'Holm' and 'BH' mark which corrections reject each pair at $\alpha = 0.05$. AUC below 0.5 means the borrowed prior is worse than chance.

pair	**AUC**	**raw p**	**Holm**	**BH**
claude-code→codex	0.394	0.0001		
codex→opencode	0.416	0.0001		
codex→cline	0.401	0.0002		
opencode→claude-code	0.469	0.0062		
cline→opencode	0.534	0.0094		
cline→claude-code	0.471	0.0184		
codex→claude-code	0.489	0.2258		
claude-code→opencode	0.486	0.3418		
claude-code→cline	0.470	0.3468		
opencode→cline	0.519	0.4714		
cline→codex	0.460	0.5365		
opencode→codex	0.458	0.5365		

nominally significant and three survive Holm. We therefore report it as *BH-dependent*: it is the only positive-direction comparison in the family, it is the weakest of the six BH rejections, and the more conservative procedure that controls the probability of *any* false rejection does not retain it.

Five of the twelve pairs are significantly below chance under the session-level unit (six under the action-level unit), which reframes the headline. "Cross- agent failure priors do not transfer" understates the result: where this corpus detects a cross-harness effect at all, the effect is usually that a borrowed prior makes the target's failures *less* predictable. The significance of that anti-prediction is what survives correction; the single positive result does not.

Stored in `research/e1_multiplicity.json`; the Fisher combined test over the twelve gives $p \approx 5 \times 10^{-10}$ but is optimistic here because the comparisons share harnesses, so we do not rely on it.

Two further pairs have point estimates above 0.5 but intervals that include it, and are therefore not established:

- **opencode→cline** (0.519, 95% CI [0.468, 0.573]): 74.3% of cline's action kinds appear in opencode's vocabulary, yet the interval spans chance.
- **opencode→codex** (0.503, 95% CI [0.179, 0.555]): three target sessions give an interval too wide to interpret. This pair is the clearest case of the point-estimate/bootstrap-mean divergence flagged in Table 2: its point estimate is 0.503 but its bootstrap mean is 0.458, because the resampling distribution for a 3-session target is heavily skewed. Where the two disagree, the bootstrap mean and the interval are the ones to read.

The weakest point estimate is **codex→cline** (0.399, CI [0.359, 0.448]) — and with the multiplicity correction applied it is no longer merely the weakest, it is one of the three comparisons that survive Holm as significantly *below* chance. Codex has only 3 sessions and 662 actions, so its per-kind estimates are sparse; sparse estimates that are systematically wrong in one direction are exactly what produces a reliably anti-predictive prior.

4.3 In-corpus baselines

In-corpus AUC (leave-one-session-out, marker_error label, session bootstrap):

- claude-code: 0.627 [0.606, 0.656] (42 sessions, 14,457 actions)
- cline: 0.751 [0.690, 0.806] (69 sessions, 12,354 actions)
- codex: 0.461 [0.341, 0.508] (3 sessions, 662 actions)
- opencode: 0.680 [0.642, 0.718] (288 sessions, 48,460 actions)

Codex's in-corpus AUC (0.461) is at chance, and its interval is by far the widest: 3 sessions provide too little data for a per-kind prior. Two of the three other harnesses' priors exceed Codex's own baseline on point estimate (opencode→codex: 0.503; cline→codex: 0.513) while one does not (claude-code→codex: 0.442). All three intervals overlap Codex's own, so none of these comparisons is individually established. Pooled, they look like a **cold-start** effect — a borrowed prior from a larger corpus beating a thin local history — but three overlapping intervals do not establish it. This comparison is not affected by the leakage that invalidated our original cold-start curve (codex's own baseline is leave-one-session-out and the borrowed priors come from other harnesses, so both arms are clean); it remains unestablished for the ordinary reason that the intervals overlap. The corrected within-target experiment in Section 4.5 is what speaks to cold-start directly, and its answer is that borrowing never helps — the borrowed prior is below chance at every history size, and local data beats it as soon as there is any.

4.4 Coverage correlates with transfer success

Figure 2 plots transfer AUC against coverage — the fraction of target action kinds the source has observed. Coverage correlates positively with transfer AUC (Pearson $r = 0.43$, $n = 12$, $p > 0.05$), but the relationship is **not** a threshold. Two pairs with coverage above 0.46 still fall below chance: claude-code→cline (coverage 0.645, AUC 0.472) and claude-code→codex (coverage 0.470, AUC 0.442). Conversely the best transfer in the matrix, cline→opencode (AUC 0.534), has coverage of only 0.375.

The pairs that fail worst (codex→cline at 0.399, codex→opencode at 0.415) have two of the lowest coverage values in the matrix (0.353 and 0.172), which invites the reading that low coverage predicts failure. The matrix refutes that reading twice over. First, monotonicity fails within a single source: codex→claude-code has *lower* coverage than codex→cline (0.241 against 0.353) yet a *higher* AUC (0.489 against 0.399) — moving coverage down moved performance up. Second, the matrix's best transfer contradicts the ordering it would need: cline→opencode achieves the highest AUC in the matrix (0.534) at coverage 0.375, only 0.022 above the failing codex→cline pair, and the two differ by 0.135 in AUC. Coverage therefore does not order these pairs, and we do not claim sufficiency in either direction. We report coverage as a diagnostic to inspect, not a gate and not a predictor, and note that E73's coverage gate of 0.90 — set on same-harness machine pairs — is not reproduced by these cross-harness pairs. Because no pair shows positive transfer once the twelve comparisons are corrected for multiplicity (Section 4.2), this correlation rests on twelve point estimates of which eleven are indistinguishable from chance and three are significantly below it; a pattern in which the two worst pairs also happen to have low coverage is the pattern expected by construction in such a sample. We present the correlation as exploratory, do not interpret it causally, and would require a larger harness set before treating it as a scaling relationship.

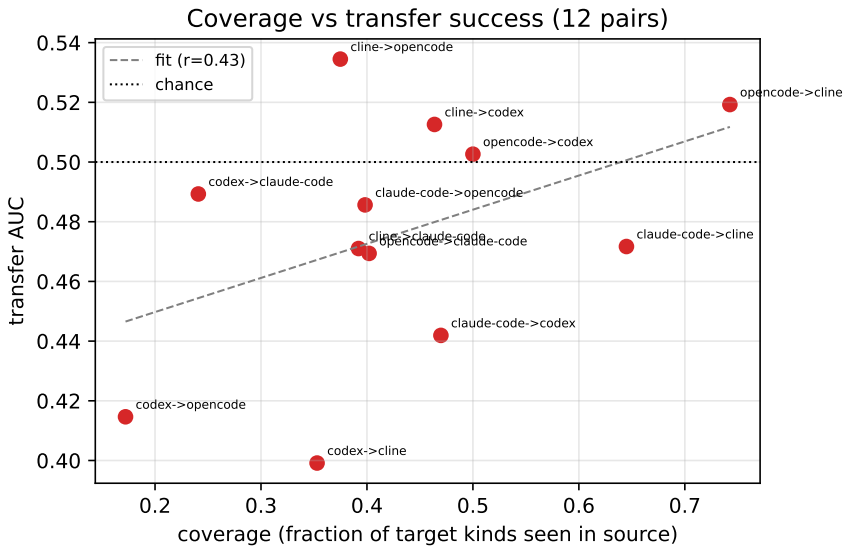


Figure 2: Coverage vs. transfer AUC. Each point is a source→target pair. Coverage is the fraction of target action kinds present in the source vocabulary. Higher coverage predicts higher transfer success.

4.5 Cold-start: a withdrawn curve, and a corrected experiment

First, the withdrawal. Our originally reported curve — 0.469 (borrowed only) rising to 0.720 (own history), with a crossover at $n = 2$ — is contaminated and is not a result. It was produced by building the local prior from target sessions and then scoring that prior on *the same* target sessions: `own_only` (n=42) used all 42 claude-code sessions to build and all 42 to score, and `mixed_n`N scored on all of `tgt` including the `tgt[:N]` it was built from. The `transferred_only` point was the only uncontaminated one.

As originally reported (withdrawn, retained as an error record): 0.469 [0.445, 0.490] for the borrowed prior alone, 0.720 [0.700, 0.743] for own history, mixed exceeding borrowed-only at $n = 2$ (0.598 [0.562, 0.635]).

Second, the corrected experiment. We re-ran it with the target harness's 42 sessions split once, seeded, into a 30-session training pool and a 12-session held-out test set (6,132 test actions). Every arm is scored on the same held-out set; N is swept over the training pool only, so no test session ever contributes to a prior; and the comparison is paired — one resample of test sessions, both arms scored on it, difference recorded, 2,000 times — so the interval is on the difference. Four arms: local-only (first N training sessions), borrowed-only (`opencode`, no target data), mixed (local + borrowed counts), and pooled-others (all non-target harnesses).

Three things follow, and the third overturns our original framing.

1. **Local history pays off almost immediately.** Two training sessions are enough to reach 0.600, and by 20 sessions the local prior sits at 0.656.

2. **The borrowed prior is below chance at every N, and never improves.** It is fixed at 0.483 because it is fixed *in fact* — no amount of target data enters it. This is the same anti-predictive pattern Section 4.2 finds, now visible within a single target.

3. **There is no cold-start regime in the sense we claimed.** Our original story was that

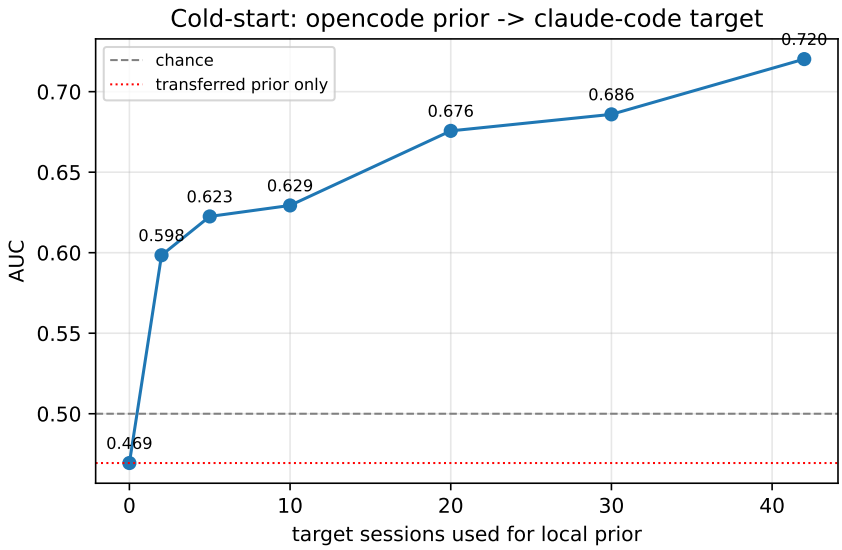


Figure 3: WITHDRAWN — shown only to document the error. This is the original cold-start curve. The `own_only` and `mixed_n`N points built their priors from sessions contained in the test set, so their apparent advantage over the borrowed-only point is leakage, not local history. Do not read any part of it as a result; Table 3 supersedes it.

> a borrowed prior helps while local history is thin and loses once it matures. The corrected curve has no such region: the borrowed prior never helps, at any N, and local data beats it as soon as there is any. The claim we can defend is weaker in mechanism and stronger in force — *borrowed priors do not help even where we expected them to help most.*

Mixed and pooled-others track local-only closely and never separate from it: at $N = 10$ local-only (0.652) is in fact *above* mixed (0.629), and at $N = 2$ the difference is within noise. Adding borrowed counts to local data buys nothing measurable here.

Stored in `research/e1_coldstart_paired.json`. This experiment is what the original should have been: the defect was leakage, not the choice of arms, and the leakage is what produced both the apparent improvement and the apparent crossover.

5 Discussion

What transfers. Almost nothing transfers across harnesses, and the exception does not survive correction. Only one of twelve pairs has a point estimate above chance with an interval excluding it (cline→opencode, 0.534), and it fails Holm (Section 4.2); it survives only the less conservative Benjamini–Hochberg. Its mechanism, insofar as we can read one, is ordinary: different harnesses share tool vocabularies (Bash, Read, Edit, Write, Grep), and where two harnesses' kind distributions happen to align the per-kind rate carries over. We cannot identify a property of a pair that predicts which case it will be — coverage, the most plausible candidate, is exploratory at best on twelve points.

What anti-transfers. The reliable effect in this corpus runs the other way. Three pairs are significantly *below* chance after family-wise correction (0.394, 0.401, 0.416), all involving Codex. Codex's corpus is 3 sessions and 662 actions, so its per-kind rate estimates are sparse; a sparse

Table 3: Cold-start, corrected. Held-out test set is fixed across every row. Intervals are paired session-level bootstrap (2,000 resamples) on the difference from the borrowed-only arm. Arm A is undefined at $N = 0$: there is no local history to build a prior from. 15 of the 18 differences exclude zero.

N	A local-only	B borrowed-only	C mixed	D pooled-others	A − B
0	—	0.483	0.483	0.482	—
2	0.600	0.483	0.608	0.605	+0.114 [+0.040, +0.153]
5	0.625	0.483	0.627	0.625	+0.138 [+0.081, +0.168]
10	0.652	0.483	0.629	0.626	+0.167 [+0.123, +0.199]
20	0.656	0.483	0.667	0.652	+0.172 [+0.123, +0.219]
30	0.649	0.483	0.667	0.666	+0.166 [+0.118, +0.221]

estimate that is systematically wrong in one direction produces a prior that is reliably worse than guessing. We flag this as an observation rather than a mechanism — we have one small harness, not a study of small harnesses — but it is the only cross-harness effect here that survives correction, and it is the result the title should be read against: the danger is not that borrowed priors fail to help, it is that they can mislead.

What does not transfer. Codex's 3-session corpus is too small for its own in-corpus prior to be useful (AUC 0.461), and other harnesses' priors beat it on point estimate. We previously read this as evidence of a cold-start effect; that reading depended on a within-target curve which we have since withdrawn for train/test leakage (Section 4.5), and the present comparison is unestablished for the ordinary reason that the intervals overlap. Across all twelve pairs the borrowed prior never reaches the target's own mature baseline.

Label drift. The four harnesses use different failure labels (harness flags vs. real exit codes vs. output markers). This introduces a small, uncontrolled confound into every cross-harness number. The direction of the bias is unknown but likely conservative: label noise attenuates AUC.

Scope. One user's fleet, one fortnight, four harnesses. The "cross-agent" label is four harnesses but one human. Cross-user transfer — whether a prior learned from one developer predicts another developer's failures — is untested and is the highest-priority external replication.

6 Conclusion

Cross-agent operational failure priors do not transfer, and the effect this corpus can detect is the opposite of the one we went looking for. Under session-level bootstrap — the correct inferential unit — and family-wise correction over the twelve comparisons, *no* pair shows positive transfer. The three pairs that survive Holm are significantly *below* chance (0.394, 0.401, 0.416): a prior borrowed from one harness makes another harness's failures less predictable than chance would. The single positive-direction result, cline→opencode at 0.534, survives Benjamini–Hochberg but not Holm, and we report it as BH-dependent rather than established.

This is a stronger claim than "priors do not transfer". The literature assumption worth testing is that failure statistics are a portable property of a task domain; on this corpus, where they are detectable across harnesses at all, they are actively misleading. That is a negative result with a sign, and it is the one the data supports.

The within-target experiment agrees. Having withdrawn our original cold-start curve for train/test leakage and re-run it properly — priors built only on a held-out training pool, all arms

scored on a fixed held-out test set, paired intervals — the borrowed prior sits below chance at every history size and never contributes, while two sessions of local history are already worth +0.114 AUC over it. We originally hoped to find a regime where borrowing is worth doing. There is none here: the answer to "should I borrow a failure prior from another agent?" is no, at every amount of local data from zero upward.

Three methodological points follow. First, the bootstrap unit is a first-class variable in this kind of evaluation: the action-level bootstrap reports two pairs as positive transfer where the session-level bootstrap reports none, and the only thing that changes is interval width. Clustering actions within sessions inflates apparent significance, and agent-trajectory work should resample sessions. Second, a plausible and often-assumed property of the data — that vocabulary coverage governs transfer — is at best exploratory here, since it is estimated from eleven chance-level point estimates. Third, and most transferable: the original cold-start curve looked smooth, monotone and plausible, and it was an artifact of scoring a prior on the sessions it was built from. A curve that rises with the amount of data an arm is allowed to see should be suspected of exactly that, and the check is cheap — rebuild the prior from a disjoint split and see whether the curve survives.

The honest summary is negative: failure structure is only weakly a property of the agent as opposed to the user, the repository, and the task. A single-user, single-fortnight corpus cannot separate those, and our cross-harness comparisons mostly cannot either.

Reproducibility

- Corpus: `benchmarks/behaviour/corpus-redacted.json.gz` (796 KB, 75,933 actions, targets salted).
- Extraction: `benchmarks/behaviour/extract_*.py` (Claude Code, Codex, Cline, OpenCode).
- Analysis: `research/papers/e74_replication.py` regenerates every number and figure in this paper from the corpus in one run and writes `research/e74_replication_results.json`.
- Build: `research/papers/build_all.sh`; claim check: `research/papers/verify_paper1.py`.